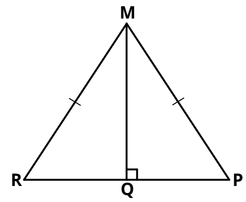
Proving triangles congruent

1 The diagram shows $\overline{MQ}$ is perpendicular to $\overline{PR}$ and $\triangle MPR$ is isosceles.

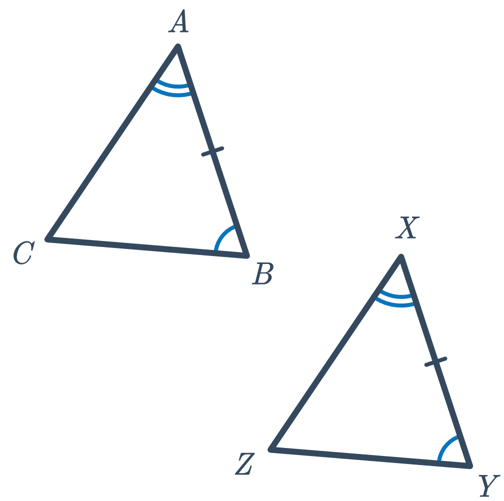


a Find $m\angle MQR$.

b Explain why $\triangle MPQ \cong \triangle MRQ$.

c Determine the congruence test that matches the argument in part b.

2 Use the information given in the diagram to complete the two-column proof that $\triangle ABC \cong \triangle XYZ$.



To prove: $\triangle ABC \cong \triangle XYZ$

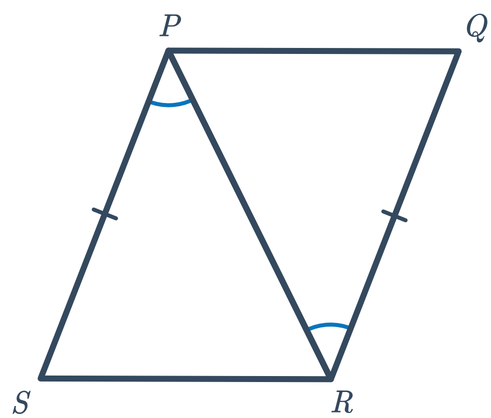
	Statements	Reasons
1.	$\angle CBA \cong \angle ZYX$	Given
2.	$\angle CAB \cong \angle ZXY$	Given
3.	$\overline{AB} \cong \overline{XY}$	Given
4.	$\triangle ABC \cong \triangle XYZ$	

3 Complete the following two-column proof t[redacted]ows that $\triangle ABC \cong \triangle XYZ$.

To prove: $\triangle ABC \cong \triangle XYZ$

	Statements	Reasons
1.	$\overline{AB} \cong \overline{XY}$	Given
2.	$\angle CBA \cong \angle ZYX$	Given
3.	$\angle CAB \cong \angle ZXY$	Given
4.	$\triangle ABC \cong \triangle XYZ$	

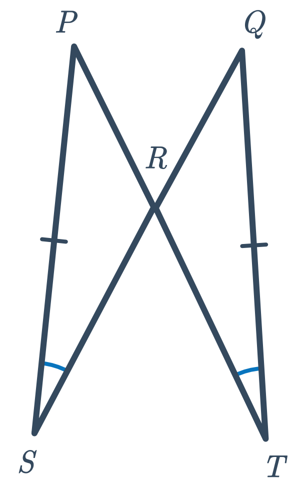
4 Use the information given in the diagram to complete the two-column proof that $\triangle PQR \cong \triangle RSP$.



To prove: $\triangle PQR \cong \triangle RSP$

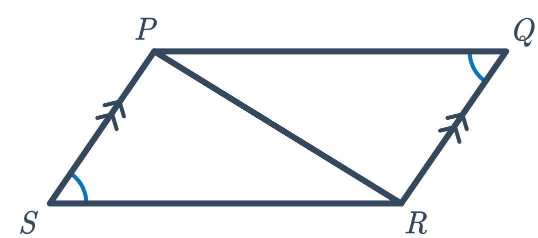
	Statements	Reasons
1.	$\overline{PS} \cong \overline{RQ}$	Given
2.	$\overline{PR} \cong \overline{RP}$	Reflexive property of congruence
3.		
4.	$\triangle PQR \cong \triangle RSP$	Side-angle-side congruence (SAS)

5 Use the information given in the diagram to complete the two-column proof that $\triangle RTQ \cong \triangle RSP$.



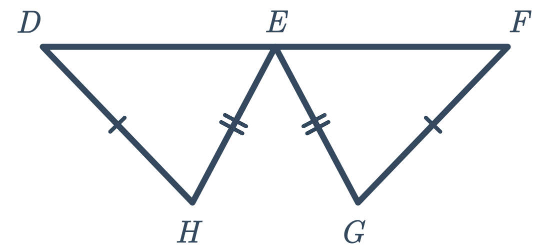
	To prove: $\triangle RTQ \cong \triangle RSP$	
	Statements	**Reasons**
1.	$\overline{PS} \cong \overline{QT}$	Given
2.	$\angle PSR \cong \angle QTR$	Given
3.		
4.	$\triangle RTQ \cong \triangle RSP$	Angle-angle-side congruence (AAS)

6 Use the information given in the diagram to complete the two-column proof that $\triangle PQR \cong \triangle RSP$.



	To prove: $\triangle PQR \cong \triangle RSP$	
	Statements	**Reasons**
1.	$\angle PSR \cong \angle RQP$	Given
2.	$\overline{PS} \parallel \overline{QR}$	Given
3.		
4.	$\overline{PR} \cong \overline{RP}$	Reflexive property of congruence
5.	$\triangle PQR \cong \triangle RSP$	Angle-angle-side congruence (AAS)

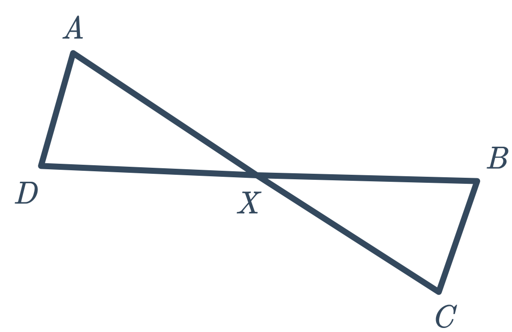
7 Use the information given in the diagram to complete the two-column proof that $\triangle DEH \cong \triangle FEG$.



To prove: $\triangle DEH \cong \triangle FEG$

	Statements	Reasons
1.	E is the midpoint of $\overline{DF}$	Given
2.	$\overline{DH} \cong \overline{FG}$	Given
3.	$\overline{EH} \cong \overline{EG}$	Given
4.		
5.	$\triangle DEH \cong \triangle FEG$	

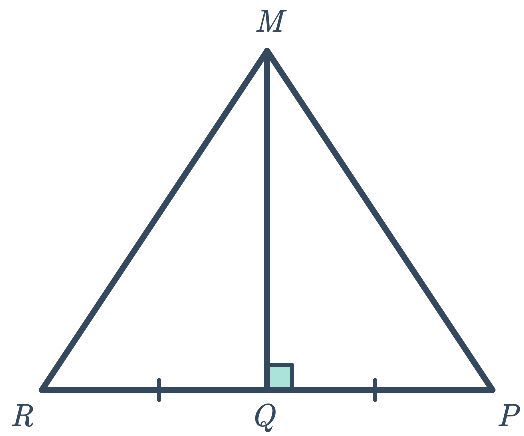
8 Use the information given in the diagram to complete the two-column proof that $\triangle AXD \cong \triangle CXB$.



To prove: $\triangle AXD \cong \triangle CXB$

	Statements	Reasons
1.	$\overline{AD} \parallel \overline{CB}$	Given
2.	$\overline{AC}$ bisects $\overline{BD}$	Given
3.		
4.		
5.	$\angle ADX \cong \angle CBX$	Alternate interior angles theorem
6.	$\triangle AXD \cong \triangle CXB$	Angle-side-angle congruence (ASA)

9 Use the information given in the diagram to complete the two-column proof that $\triangle MQR \cong \triangle MQP$.

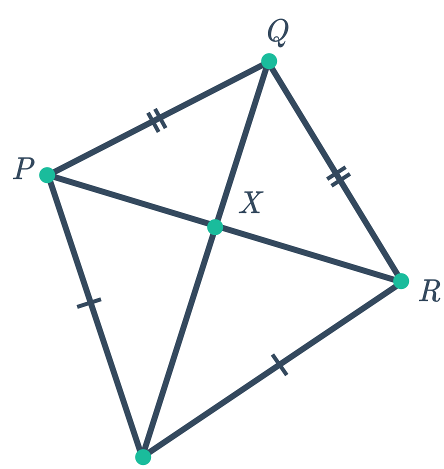


To prove: $\triangle MQR \cong \triangle MQP$

	Statements	Reasons
1.	$\overline{RQ} \cong \overline{QP}$	Given
2.	$\overline{RP} \perp \overline{MQ}$	Given
3.	$\angle MQR$ and $\angle MQP$ are right angles	Definition of perpendicular
4.		
5.	$\overline{MQ} \cong \overline{MQ}$	Reflexive property of congruence
6.	$\triangle MQR \cong \triangle MQP$	

Additional questions

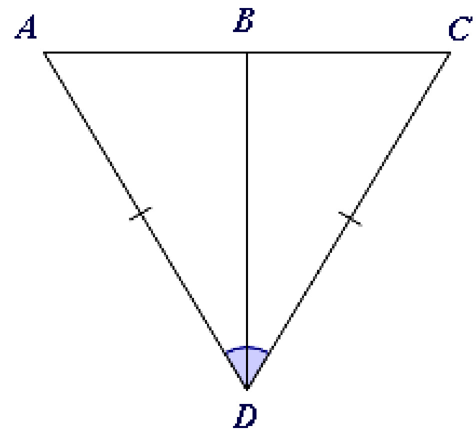
10 Use the diagram to prove that $\triangle SPQ \cong \triangle SRQ$.



11 Given:

- $\overline{AD} \cong \overline{CD}$
- $\overline{BD}$ bisects $\angle ADC$

Prove: $\triangle ABD \cong \triangle CBD$

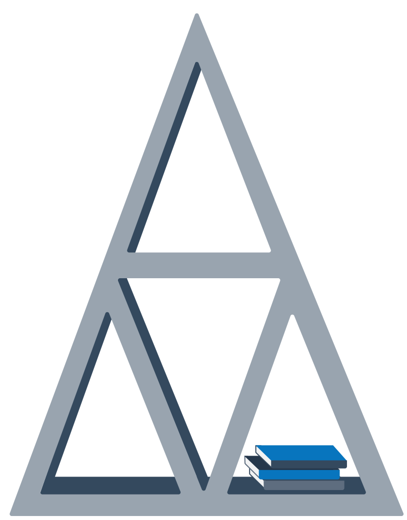


12 Given:

- $\angle A \cong \angle P$
- $\overline{AB} \cong \overline{PQ}$
- $\overline{BC} \cong \overline{QR}$

Is there enough information to conclude $\triangle ABC \cong \triangle PQR$? Draw a diagram to justify your conclusion.

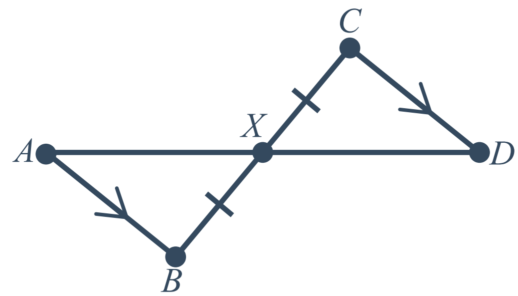
13 In construction technology class, Quentin has decided to construct a shelf with triangular openings. How should Quentin cut and assemble the pieces of wood to guarantee the triangles are congruent? Assume he will use no more than six planks to construct the shelf. Use a triangle congruence postulate to justify your solution.



14 Given:

- $\overline{CD} \parallel \overline{AB}$
- $\overline{XB} \cong \overline{XC}$

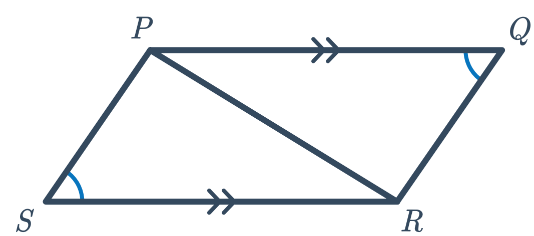
Prove: $\triangle ABX \cong \triangle DCX$



15 Given:

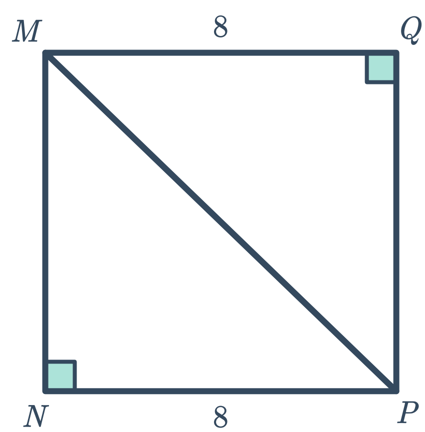
- $\overline{PQ} \parallel \overline{SR}$
- $\angle S \cong \angle Q$

Prove: $\triangle PRS \cong \triangle RPQ$

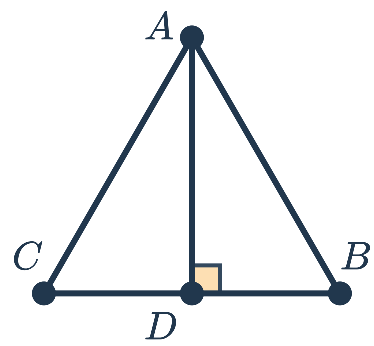


16 Explain why Angle-angle-angle (AAA) is not a valid triangle congruence theorem.

17 Use the information given in the diagram to complete the paragraph proof that $\triangle MQP \cong \triangle PNM$.
From the diagram we know that $\overline{MQ} \cong \overline{NP}$ by definition of congruence. We also know that ▯ and ▯ are right angles so $\triangle MQP$ and $\triangle PNM$ are both ▯ triangles by the ▯. Next, we know that ▯ $\cong$ ▯ by the Reflexive property. Therefore $\triangle MQP \cong \triangle PNM$ by the ▯ congruence theorem.



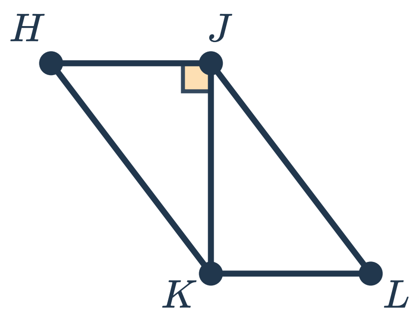
18 In the figure shown, $\triangle ABC$ is equilateral.
Prove $\triangle ACD \cong \triangle ABD$.



19 Given:

- $\overline{HJ} \parallel \overline{KL}$
- $\overline{HK} \cong \overline{LJ}$

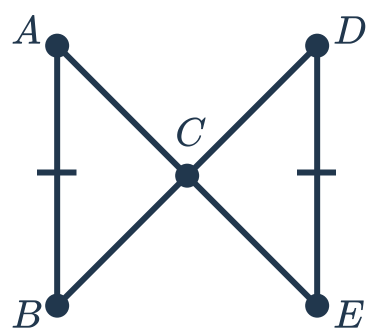
Prove: $\triangle HJK \cong \triangle LKJ$



20 Given:

- $m\angle DCE = 90^\circ$
- $\overline{AE}$ bisects $\overline{BD}$

Prove: $\triangle ABC \cong \triangle EDC$



21 A rectangle can be divided into two triangles by either of its diagonals. Explain how these two triangles satisfy the Hypotenuse-leg (HL) conditions for triangle congruence.